PYTHON • TKINTER PROJECT

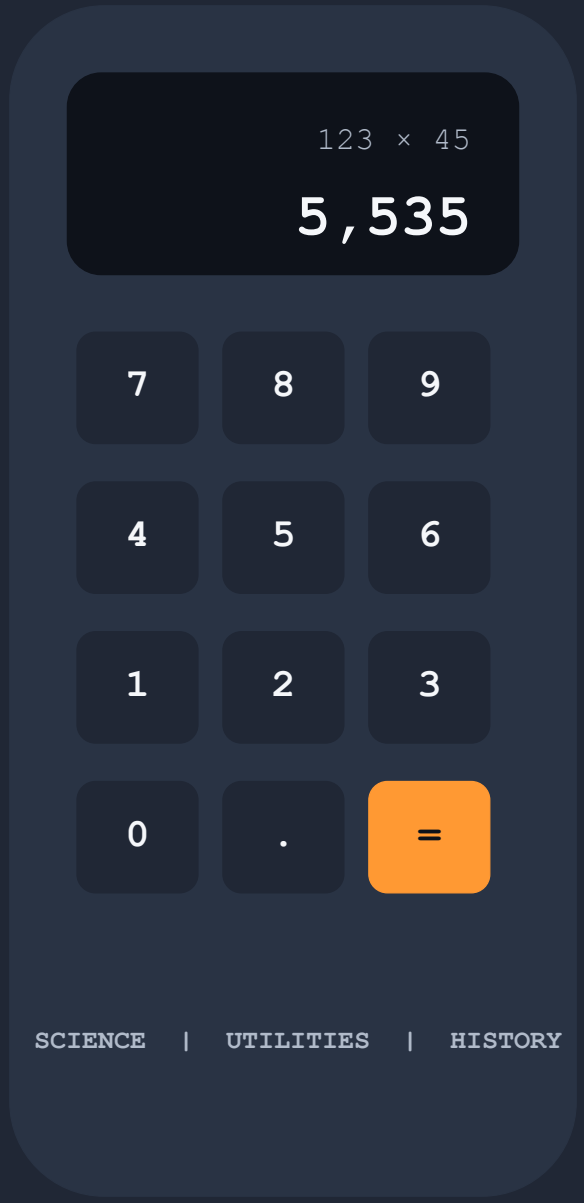
PROFESSIONAL CALCULATOR

A multi-function desktop calculator built with Python

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Project Overview

One desktop application combining everyday calculations and scientific tools.

Project Goal

- Build a practical calculator with a clean desktop interface.
- Support standard arithmetic and scientific operations.
- Add useful mini-tools in one application.

Technology Stack

- Python for application logic.
- Tkinter for the graphical user interface.
- Math and AST modules for calculations.
- SymPy for polynomial equation solving.

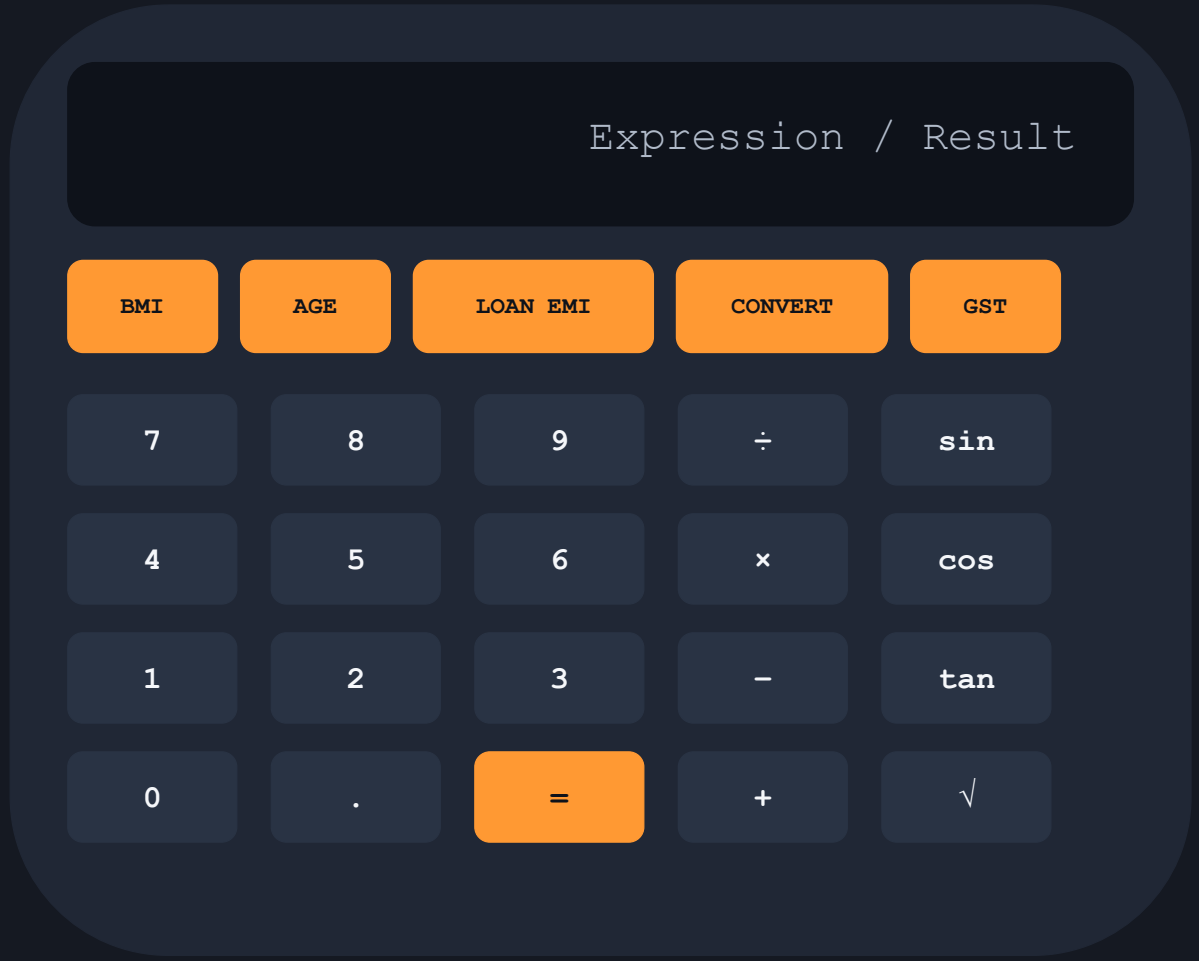
User Experience

- Dark interface with clear display and buttons.
- Utility buttons positioned below the display.
- Calculation history available inside the app.
- DEG / RAD angle mode.

Application type: Windows desktop GUI

Main Calculator Interface

The primary calculator combines a readable display, quick utility access and a scientific keypad.



Layout highlights

- Main display at the top.
- BMI, AGE, LOAN EMI, CONVERT and GST directly below it.
- Scientific and numeric keys below the utility row.

Built-in controls

- Clear and backspace actions.
- History access.
- DEG / RAD toggle and constants such as π and e.

Scientific Calculation Features

Designed for common scientific and advanced mathematical operations.

Trigonometry

- sin, cos, tan
- asin, acos, atan
- sinh, cosh, tanh

Logarithmic & powers

- logarithmic functions
- square root and powers
- exponential operations

Other operations

- factorial and percentage
- π and e constants
- DEG / RAD angle selection

Angle mode matters: DEG interprets trigonometric input in degrees; RAD interprets it in radians.

Five Built-in Utility Tools

Each tool opens its own focused input window from the buttons below the main display.

BMI

Uses height and weight to calculate BMI and show a category.

AGE

Calculates age from the entered date of birth.

LOAN EMI

Calculates monthly EMI and loan repayment details.

CONVERT

Converts values between supported measurement units.

GST

Calculates GST amount, total amount and CGST / SGST split.

Quick access: all five utility buttons sit immediately below the calculator display.

Smart Features & Calculation Flow

Beyond keypad calculations, the application includes equation solving and session history.

Equation Solver

- Polynomial equation solving using SymPy.
- Accepts equations containing x.
- Displays the calculated solution set.

Calculation History

- Stores completed calculations during the session.
- Shows previous expressions and results.
- Allows a selected history item to be reused.

Safer Evaluation

- Uses a restricted AST-based expression evaluator.
- Allows only supported numeric operations and functions.
- Avoids evaluating arbitrary Python statements.

Typical flow: Enter expression → Calculate → View result → Review or reuse from History

How the Application Works

A simple layered structure keeps the interface, calculation logic and utility tools organized.

USER INPUT — Buttons • Keyboard • Utility forms

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TKINTER GUI — Display • Scientific keypad • History • Tool windows

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CALCULATION LOGIC — Safe expression evaluation • Math functions • EMI / GST / BMI / Age / Conversion

↓

LIBRARIES — math • ast • tkinter • sympy

The GUI collects input, calculation functions process it, and results are presented back in the interface.

Testing, Outcome & Future Scope

A summary of what the current build delivers and where it can be improved next.

Current Build

- Standard and scientific calculations.
- Five utility calculators / converters.
- Equation solver and calculation history.
- Dark-themed Tkinter desktop interface.

Validation Checklist

- Check arithmetic and scientific results.
- Test DEG / RAD mode separately.
- Verify utility inputs and invalid-value handling.
- Test equation and history workflows.

Future Enhancements

- Add memory keys and customizable themes.
- Export or save calculation history.
- Add more unit categories and financial tools.
- Package as an installable desktop application.

THANK YOU

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